
Practice of Forestry - education & communication

Nine Tips to Improve Your Everyday Forest Data Analysis

Matthew B. Russell[✉]

Matthew B. Russell (russellm@umn.edu), Department of Forest Resources, University of Minnesota, 1530 Cleveland Avenue N, St. Paul, MN 55108.

Abstract

Forestry data are complex—they are collected across long time periods, at different resolutions, and include multiple types and sources. Forestry professionals are increasingly relying on new technologies to make informed decisions using these data. This article describes nine practices that forestry professionals and their organizations can take to improve their everyday forest data analysis. By integrating these practices into their work, forestry professionals can sharpen their data management and analytical skills and contribute to more effective decisions made by their organizations.

Study Implications: Forestry professionals contribute to the collection and analysis of data, but less training is provided to them in managing, organizing, and communicating data. A considerable amount of time is spent organizing and restructuring data in forestry, but these actions do not lead to immediate results that can inform decisions. Creating a workplace culture where data are high-quality and trustworthy can assist forestry professionals in using their analytical skills to address problems in areas such as forestland acquisition, carbon sequestration, and ecosystem services. If data are going to continue to drive forest management and policy decisions at the stand, landscape, or national scale, then data analysis skills need to be valued by foresters and the organizations for which they work.

Keywords: forest data, forest analytics, data visualization, data management, communication

Data literacy is defined as the ability to understand and use data effectively to inform decisions ([Mandinach and Gummer 2013](#)). Employers seek forest and natural resource management professionals that have these skills. During their coursework and training, forestry students typically complete coursework that provides them with tools to summarize data, test statistical hypotheses, and interpret their findings. Within the quantitative disciplines in forestry, exposing students to unorganized and messy data sets with missing values can contribute to problem-based and experiential learning. [Temesgen et al. \(2011\)](#) discuss the value of problem-based and experiential learning in the

training of a quantitative forestry student in the 21st century.

Forestry and natural resources data are often collected across long time periods, contain multiple resolutions (e.g., at tree, stand, and landscape scales), and integrate multiple data sources (e.g., remote sensing and on-the-ground measurements). In any data analysis project, a considerable amount of time is spent organizing and structuring these heterogeneous data so that they can be used.

The consequences of poor forest data management are grave. For example, a flood in a USDA Forest Service building can destroy original paper data files

from a long-term study (Kenefic et al. 2013), thousands of digital data files in different data formats are not readily accessible to forestry researchers (Russell et al. 2014), and insufficient quality assurance/quality control processes in a national forest inventory can lead to a failure to observe trends in hardwood log quality (Brandeis et al. 2017). The number of forest scientists decreased by 15 percent from 2002 to 2016 (McGinley et al. 2019) suggesting that in today's forest research sector, data analysis tasks are likely spread across more individuals within an organization.

This is an article for forestry professionals that use data in their daily work. It is particularly well suited to forestry professionals that use quantitative data and are in the early stages of their careers. This article does not emphasize specific analysis techniques or procedures for summarizing forestry data. Instead, nine tips are provided that focus on strategies for more effective data analysis.

Organize Data to Make It Analysis Ready

After data collection, most data sets in forestry are not in a format that enables them to be immediately analyzed. The primary reasons for this are because most people are not familiar with well-organized data structure and data are often organized for a task other than analysis, such as data entry (Wickham and Grolemund 2017). Surprisingly, the majority of the time spent by data science professionals (53 percent of it) involves collecting, labeling, cleaning, and organizing data, not analyzing and modeling it (CrowdFlower 2017).

For example, after tree diameters are remeasured in a permanent sample plot, are they entered into a new spreadsheet for data analysis? As alternatives, the diameters could be entered in a new column within an

existing spreadsheet or as a new row in the existing spreadsheet.

A number of useful strategies for organizing data in spreadsheets and tabular formats are presented in Wickham (2014) and Broman and Woo (2018). Collectively, these strategies emphasize keeping data tidy—that is, every variable is a column, every observation is a row, and each type of observational unit is contained in a table (Wickham 2014). Implementing these strategies in practice often involves creating “rectangles” of data within spreadsheets with no missing values, in addition to using short, meaningful, and consistent names for variables (Broman and Woo 2018).

In the tree diameter example, a tidy data set might include all diameter measurements in a single spreadsheet with one column for tree diameter and another for the date it was measured. Attributes of a nontidy data set might include the presence of cells with null values, not displaying clear names of variables, and mixing fonts and colors that add little value to data analysis (Figure 1).

Visualize Data First

Good analyses start with exploring the data using graphics. Often termed exploratory data analysis, visualizing raw data allows an analyst to discover patterns in the data, identify interesting observations such as outliers, assess whether data transformations are needed, and find new hypotheses to test (Tukey 1977).

Common visualizations to start with include scatterplots, histograms, box plots, and violin plots. Using a variety of visualizations allows an analyst to investigate data that contain both continuous and categorical variables. For example, consider a data set of 333 cedar elms (*Ulmus crassifolia* Nutt.) measured in

(a) Non-tidy			
PLOT 1		DBH_2015	DBH_2020
TREE	1-ACSA	7.8	8.7
	2-ACSA	8.8	9.5
	3-BEAG	10.4	11.2
PLOT 2			
TREE	1-ACSA	6.7	7.1
	2-FAGR	8.1	8.9
	3-FAGR	8.4	9.1

PLOT	TREE	SPECIES	YEAR	DBH
1	1	ACSA	2015	7.8
1	2	ACSA	2015	8.8
1	3	BEAG	2015	10.4
2	1	ACSA	2015	6.7
2	2	FAGR	2015	8.1
2	3	FAGR	2015	8.4
1	1	ACSA	2020	8.7
1	2	ACSA	2020	9.5
1	3	BEAG	2020	11.2
2	1	ACSA	2020	7.1
2	2	FAGR	2020	8.9
2	3	FAGR	2020	9.1

Figure 1. Two examples of a nontidy (a) and tidy (b) data set used after collecting tree information from a permanent sample plot. Although both contain the same information, the tidy example facilitates a straightforward analysis.

the Urban Forest Inventory and Analysis program in Austin, TX (File S1, S2). By exploring the continuous variables with these graphics, we see a positive relationship between tree diameter and height and a distribution of heights that are right skewed with many short trees and few tall trees. After analyzing the data by tree crown class, a categorical variable, we see that open-grown trees are relatively short, dominant trees are tall, and codominant trees are both short and tall (Figure 2). A deeper investigation can look into the data's interesting observations, such as whether the 40-inch tree is a data entry error or if it was actually observed in the field.

Use Standardized Routines for Analyzing Forest Data

Many steps in the forestry professional's data analysis workflow involve performing the same task but on a different data set. These tasks may be performed while working with a spreadsheet, writing code, and/or processing data to be analyzed. When performing these routines, work smarter and not harder. By using standardized data analysis routines that are performed iteratively, you will save time and effort, minimize potential data analysis errors, and provide consistent results.

As an example, imagine we conducted a forest inventory across several stands using a stratified random sample design. Our variable of interest is the basal area per acre within each stand. The approach to summarize these data is well documented in many forest mensuration textbooks (e.g., Burkhardt et al. 2019) but requires multiple calculations. Using a standardized routine, a user may load the data into R software (R Core Team 2019) and use the suite

of functions compiled by Wolken and Rodman (2019) to summarize the inventory data. The result is an estimate of the mean basal area in each stand, in addition to confidence limits and an accompanying visualization showing the distribution of basal area across the inventory.

Before developing a new routine, ask yourself if someone else in the forestry profession has already done it for you. A number of routines have likely been standardized by your coworkers, colleagues, and other professionals. The R program, the most popular statistical software being taught in undergraduate curricula today, contains tens of thousands of packages, or collections of functions and data sets developed by R users. There are several R packages developed by forestry professionals that perform many of the standardized routines that may be of interest to you (Table 1).

Document and Share Code

Increasingly in the natural sciences, analysts are moving away from using “click-and-calculate” software packages to ones in which analysts write and implement customized scripts (Borregaard and Hart 2016). A good practice is to write plentiful comments within code or keep a notebook to record data analysis processes that both worked and failed. Write comments that explain the “why” of your code, not the “what” and the “how” (Wickham and Grolemund 2017). For example, a useful comment prior to a block of code that calculates the volume of trees from a tree list might read: “calculating the merchantable volume of trees based on USDA Forest Service volume equations to later summarize stand-level volume across

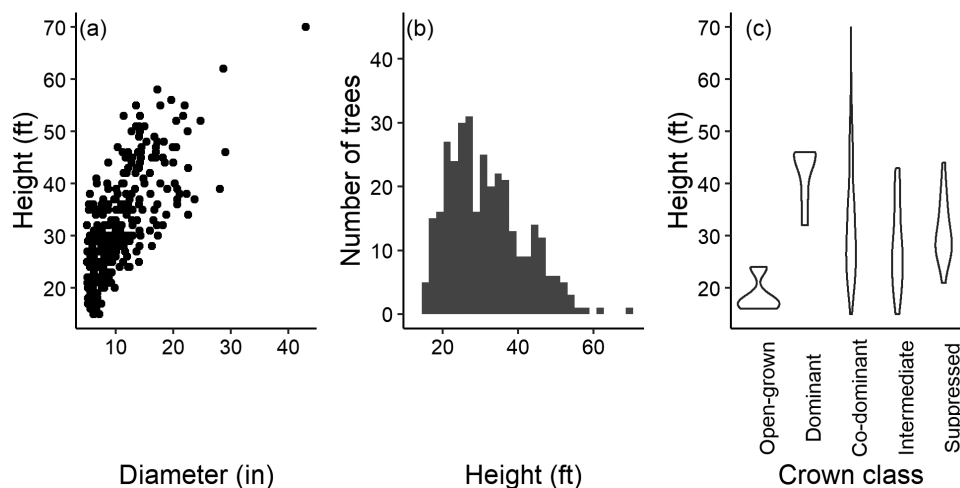


Figure 2. Some common visualizations to explore forestry data include scatterplots (a), histograms (b), and violin plots (c), a type of visualization that is similar to a box plot but also shows a kernel probability density of the data. Data are from 333 cedar elm trees measured in the Urban Forest Inventory and Analysis program in Austin, TX.

Table 1. A sampler of R packages useful in forestry applications.

R Package	Citation	Description
burnr	Malevich et al. (2018)	Contains functions that analyze forest fire history data.
caliver	Vitolo et al. (2018)	Performs calibration and verification of gridded wildfire danger models.
canopyLazR	Kamoske et al. (2019)	Estimates leaf area density, leaf area index, and forest structural attributes from airborne lidar point clouds.
lidR	Roussel and Auty (2020)	Manipulates and visualizes airborne lidar data for forestry applications.
ForestFit	Teimouri et al. (2020)	Computes statistical models for plant size distributions (e.g., tree diameter distributions).
ForestGapR	Silva et al. (2019)	Computes canopy gap statistics for tropical forests with data from airborne laser scans.
forestr	Atkins et al. (2020)	Calculates forest and canopy structure metrics from multiple data forms, including two- and three-dimensional terrestrial lidar data.
forestsamplr	Wolken and Rodman (2019)	Creates summary statistics for common forest inventory sampling designs, including simple random, stratified, systematic, cluster, Poisson, two-stage, and 3P sampling.
rFIA	Stanke et al. (2020)	Queries and analyzes the USDA Forest Service Forest Inventory and Analysis database and estimates a variety of forest attributes such as volume, biomass, and carbon stocks.
vegan	Oksanen et al. (2019)	Contains functions for performing various community ecology analyses including diversity analysis and ordination methods.

species.” Poor, useless comments might include “calculating volume using tree diameter and height” (the use of the diameter and height variables should be clear within the block of code) and “calculating volume using a Demareeschalk taper equation” (explains the “how” behind the data analysis but not the “why”).

Being more transparent in chronicling your analysis also increases the chances that it becomes reproducible. Integrating the concept of reproducibility into your work sends the signal that your analysis is high quality, trustworthy, and transparent ([Sandve et al. 2013](#)). Version control systems such as Git (<https://git-scm.com/>) and GitHub (<https://github.com/>) promote open collaboration with your colleagues, allow you to revert to working on previous versions of code, and contribute to reproducibility ([Blischak et al. 2016](#)).

Automate Data Analysis Workflow

Automation can be defined as employing automatic processes and systems for collecting, analyzing, and archiving data. A simple way to incorporate automation into your data analysis workflow is to use validation to avoid data entry errors. An example for continuous variables includes specifying a range of

values that a variable can take, such as tree height containing a value between 0 and 120 feet. An example for categorical variables includes listing potential values in a drop-down list, such as “open-grown”, “dominant”, “codominant”, “intermediate”, and “suppressed” to represent tree crown class. A number of data validation features are available in commonly used spreadsheet software like Excel ([Microsoft Corporation 2020](#)).

Organizations with large volumes of data collected across varying sources will benefit from investing in extract, transform, and load (ETL) processes to automate their data workflow ([Tiwari 2016](#)). These ETL processes use heterogeneous data (those in a variety of shapes and formats) and turn them into “homogenized” data that can be readily analyzed. Many of these ETL processes have traditionally been done through mainframe computers; however, with the introduction of business intelligence software, a more holistic integration of current and historical data is readily available. An example application could be a large forest products company that receives hundreds of deliveries of wood on log trucks to its mill on a daily basis. An analyst could rely on an ETL system to query historical and current data from multiple sources (e.g., log truck wait times and mill operation

capacities) to identify stressful periods of mill production that may slow production. These ETL processes can increase an analyst's productivity because they automate processes that integrate data and do not require writing scripts every time a new analysis is conducted.

Ask if a Table, Figure, or Interactive Visual Is Best

Good analyses end with summarizing the data using graphics. Creating tables generally requires less effort than designing figures and graphics. As the statistician Andrew Gelman (2011, p. 37) warns, "Anyone can run a regression or an ANOVA (analysis of variance)! Regression and ANOVA are easy. Graphics are hard."

It is best to use tables under three scenarios:

1. To show precise numbers or text. As an example, when a forestry professional purchases stumpage, she would like to know that the aspen pulpwood price is exactly \$34 per cord, not about \$30.
2. To compare numbers, such as values for different species or contrasting experimental treatments. For example, a table might show that the price of aspen pulpwood is \$34, whereas birch pulpwood is \$26 per cord.
3. To present different kinds of information with different units of measure. For example, a table might show that the price of aspen pulpwood is \$34 per cord, whereas pine sawtimber is \$170 per thousand board feet.

Figures are best used under two scenarios:

1. To display trends or relationships between different values through time. As an example, figures can be used to show how the price of aspen pulpwood has changed over the last 20 years or how the price of aspen pulpwood compares through time with other species.
2. To illustrate trends when the raw values are less important. Common examples used in forestry include photographs, maps, graphical abstracts, and other illustrations.

After summarizing the cedar elm data presented previously, the choice of a figure or table may be up to the user. The table is ideal if we need to compare specific values across crown classes (Table 2). The figure is

Table 2. A table that summarizes the height of cedar elm trees in Austin, TX.

Crown Class	Number of Trees	Mean (ft)	Standard Error (ft)
Open grown	4	18.8	1.8
Dominant	6	42.0	2.4
Codominant	269	32.0	0.6
Intermediate	36	28.2	1.3
Suppressed	18	30.9	1.4

more effective if we want to examine the general trends across crown classes (Figure 3).

Tables and figures are static representations of data, whereas interactive data visualizations are becoming much more common (and easier to create with new software). Interactive data visualizations are best used with data sets that contain a large number of observations and numerous variables. Examples include ArcGIS StoryMaps (e.g., [USDA Forest Service 2020](#)), Shiny web applications developed in the R programming language (e.g., [Minnesota Department of Natural Resources 2020](#)), and customized web/mobile applications such as My City's Trees ([Texas A&M Forest Service 2019](#)).

Communicate Data and Results Clearly

Consider your audience when presenting your data and the interesting things about it. Are there some audiences that prefer one kind of visualization over another? Charts, figures, and infographics are plentiful in today's world but are not always effective in conveying the messages we want to send. In some cases, visualizations can be deceiving (Cairo 2019).

As an example, it is not a good practice to include a table with 10 columns and 20 rows of data in a presentation on a slide that you show for 10 seconds. It is simply not enough time for anyone to digest that volume of information. Instead, use a table in a presentation that only shows the key results. The table with 20 rows of data might belong in a written report. When using tables and figures in a presentation, use the rule to design the slide for the back of the room.

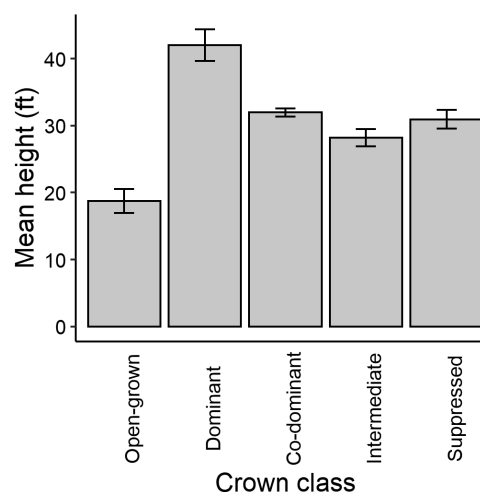


Figure 3. A figure that summarizes the height of cedar elm trees in Austin, TX ($\pm$ one standard error).

Make your fonts large enough to be seen by everyone in the room and use colors that effectively get your message across. As noted by [Harrower and Brewer \(2003\)](#), choose colors for your figures that are appropriate for your data and are friendly to color-blind individuals. Additional good practices on creating effective data visualizations can be found in [Evergreen \(2013\)](#) and [Fandrey \(2017\)](#).

The charts and figures you design for your colleagues and technicians in your organization (i.e., those with forestry and natural resource backgrounds) should not likely be the same ones you share with the general public. Assume that anyone that does not have a forestry degree cannot define what basal area is, much less comprehend the units of square feet per acre. Present graphs to the general public that use easy-to-understand units, are free from jargon, and are supplemented with text to convey key messages. As an example with the cedar elm data, in lieu of summarizing the cedar elm data by tree crown class (i.e., [Figure 3](#)), present it in terms of how much sunlight a tree receives, a concept understood by all audiences ([Figure 4](#)).

Follow the 3-2-1 Rule for Data Backups

Forestry data are collected and stored in both digital and analog formats and are often managed by several individuals. Despite advances in data storage, risks of data loss and file corruption still remain across many organizations. A unique feature of forestry research is that data are accumulated across long time periods. Without proper data backups, long-term forestry

research data can become lost or out of date. [Kenefic et al. \(2013\)](#) report several misfortunes of data loss in forestry research. For example, [Kenefic et al. \(2013\)](#) describe cases where eminent forestry researchers left behind important long-term data sets with no instructions on the content of the data, a flood occurred in a Forest Service building in Maine and destroyed paper files containing records from a long-term silvicultural trial, and files sent to a central data record center were lost with no local copies to serve as a backup.

The 3-2-1 rule for data backups stipulates keeping three copies of your data on two different types of storage or media with at least one copy off-site. Keeping three copies of your data allows you to be working continually on one copy and backing up two other versions. Two backup copies should be stored on two different types of storage (e.g., on an external hard drive and in a cloud environment). Keeping one copy of the data off-site such as in a different physical location or the cloud can eliminate any losses of data from disasters such as floods, fires, or building collapses.

Have a Data Management Plan

Adequate management of data is an essential component of any organization, but this task is often disregarded because of limited resources and the short-term nature of many projects ([Burton 2006](#)). For those organizations that invest in sound data management practices, the number of databases, file formats, and storage devices is overwhelming. In particular, with forestry data the abundance of data management systems that are available can create problems with using and communicating data. Improper analysis and communication of data can lead to managers making poor decisions ([Robinson and Hamann 2011](#)).

Data management plans (DMPs) outline how data will be administered throughout their life. Good DMPs are “living documents” that are regularly reviewed and revised as needed ([Michener 2015](#)). Most DMPs include sections on the types of data, metadata (data that provides information about aspects of the data), the roles and responsibilities of individuals tasked with managing data, a timeline for data storage, and how the data will be shared within and outside the organization.

An important component of any data management plan will include software and tools. Relational database management systems are useful systems for archiving tabular forestry data that are collected in a hierarchical format (e.g., trees measured on plots

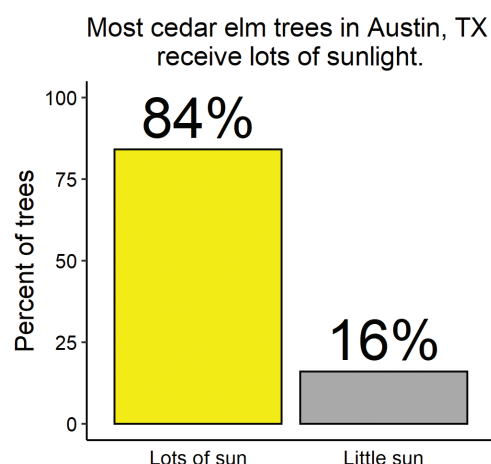


Figure 4. A figure for the general public that summarizes cedar elms trees in Austin, TX, based on the percent that receive lots of sun (open-grown, dominant, and codominant trees) and little sun (intermediate and suppressed trees).

found in stands), are long-term in their design, and can allow a user to query the system for the variables of interest (Russell et al. 2014). Additional data management systems that can integrate various information formats are often marketed as data management platforms, data integration tools, and/or business analytics programs.

Conclusion

Before the analysis begins, taking steps such as organizing data efficiently, visualizing it, and implementing standardized routines can streamline the data analysis process. When the analysis ends, clearly communicating data and results and implementing good data management practices can allow them to be used effectively in the future and will lend credibility to your work. Your audience will influence how you display data and results—graphics for a technical audience should not be the same as for the public. Dedicating time to improving your forest data analysis skills will enhance the credibility of your work and will pay you dividends when your work is used to solve complex problems.

Supplementary Materials

Supplementary data are available at *Journal of Forestry* online. **Supplement 1.** Data file (.csv) for 333 cedar elm trees measured in the Urban Forest Inventory and Analysis program in Austin, TX.

Supplement 2. R Markdown script (.Rmd) that analyzes and visualizes cedar elm data.

Supplement 3. Forestry-specific R packages listed on the Comprehensive R Archive Network (<https://cran.r-project.org/>; accessed 4 May 2020).

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